# UNIT 8

# FORCES AND THEIR EFFECTS

## Learning objectives:

- Define force and its effects on objects, such as changing motion or shape.
- Differentiate between contact and non-contact forces with examples.
- Understand friction and air resistance, and their roles in everyday life.

## 8.1 Forces

A force is a push or pull that can make an object move, stop, or change its direction. Forces are everywhere in our daily lives.

**For example:** When you push a door to open it, you are applying a force. Similarly, gravity pulls objects toward the ground. Forces are measured in units called **Newtons (N)**.

**Forces can:**

- Change the shape of an object.
- Change the direction of a moving object.
- Increase or decrease the speed of an object.

### Types of Forces:

1. **Contact Forces**: They require direct contact between objects, e.g., friction.

2. **Non-Contact Forces**: They act over a distance, e.g., gravitational force.

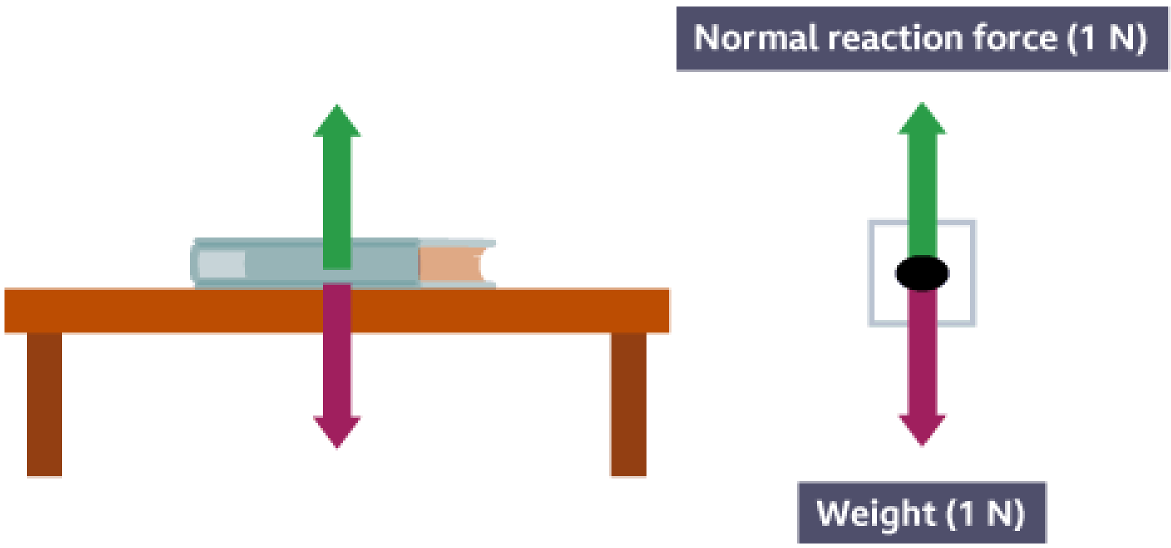
## 8.2 The Magnitude and Direction of a Force

The **magnitude** of a force refers to how strong the force is, while the **direction** indicates where the force is applied. For example:

- If you push a car harder, the force increases (greater magnitude).
- If you push it toward the left, the car moves left (direction).

Forces are represented using **arrows**:

- The length of the arrow shows the magnitude.
- The arrow's direction shows the force's direction.

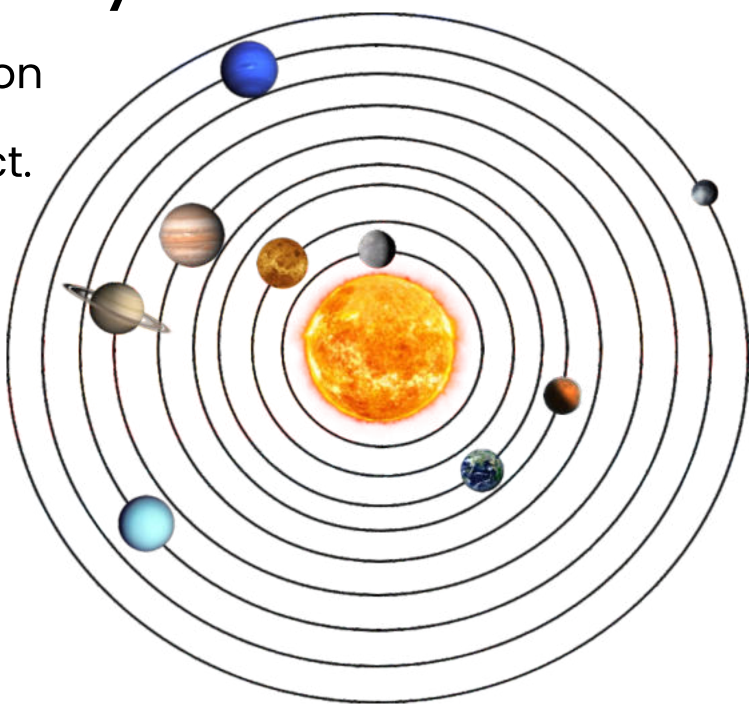


## 8.3 Gravitational Force

Gravitational force is a **non-contact force** that pulls objects toward the centre of the Earth. This is why objects fall when dropped. Gravity also keeps planets in orbit around the Sun.

❖ **Key points about gravity:**

- The force depends on the mass of the object.
- The larger the mass, the stronger the gravitational pull.
- Without gravity, we would float in space!



## 8.4 Frictional Force

Frictional force occurs when two surfaces rub against each other. It always opposes motion. For example, friction slows down a moving car when the brakes are applied.

**Types of Friction:**

**1. Static Friction:** It prevents objects from starting to move.

Example: A book on a table.

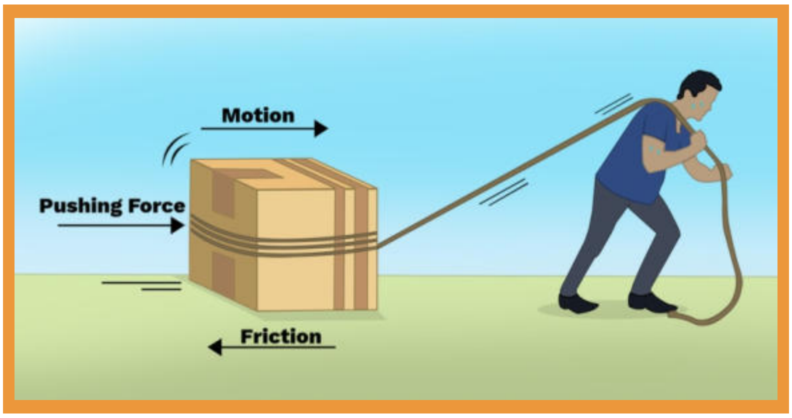
**2. Rolling friction:** The friction when an objects rolls.

Example: A ball rolling on the ground.

**Uses of friction:**

1. The friction prevents slipping while walking.

Friction enables safe braking in vehicles.




## 8.5 Air Resistance

Air resistance is a type of frictional force that acts on objects moving through the air. It slows down the motion of objects. **For example:**

- Parachutes slow down due to air resistance.
- Objects with larger surface areas experience more air resistance.

Air resistance is why lightweight objects like feathers fall more slowly than heavier objects.

## EXERCISES

### Fill in the blanks.

**1.** A force is a ______________ or a ______________.

**2.** Forces are measured in units called ______________.

**3.** The length of an arrow in a force diagram represents the ______________ of the force.

**4.** Gravitational force pulls objects toward the ______________ of the Earth.

**5.** Frictional force always ______________ motion.

**6.** Air resistance is a type of ______________ force.

**7.** ______________ friction prevents objects from starting to move.

**8.** Without ______________, objects would float in space.

## Tick(✓) for True and cross (x) for False statements

**1.** A force can only pull an object.

**2.** Magnitude shows how strong a force is.

**3.** Gravity is a type of contact force.

**4.** Friction helps us walk.

**5.** Air resistance is greater for objects with smaller surface areas.

**6.** Gravitational force keeps planets in orbit.

**7.** Friction always helps in every situation.

**8.** The direction of a force affects how an object moves.

## Choose the correct answer

**1. A force is measured in which unit?**

a) Metres b) Kilograms c) Newtons d) Joules

**2. How is the direction of a force shown?**

a) By a number b) By an arrow c) By a mass d) By a speed

**3. Gravity is an example of what kind of force?**

a) Contact force b) Non-contact force

c) Frictional force d) Magnetic force

**4. Air resistance acts in which direction relative to motion?**

a) Same direction b) Opposite direction

c) Neutral direction d) Random direction

**5.Which force works against motion?**

a) Gravity b) Friction c) Magnetic force d) Applied force

**6. What helps a parachute slow down its fall?**

a) Gravitational force b) Frictional force

c) Air resistance d) Magnetic force

**7. The Earth's pull on objects is due to which force?**

a) Friction b) Magnitude c) Gravity d) Direction

**8. Friction is greatest when surfaces are:**

a) Smooth b) Rough c) Wet d) Large

## Answer the Following Questions:

### Short-Answer Questions

**1. How does air resistance help parachute work?**

**2. Define gravitational force.**

**3. How does air resistance affect motion?**

### Long-Answer Questions

**1. What is a force? Explain the effects, types and unit of forces?**

**2. What is friction? Describe its types and work?**

**3. How does air resistance help parachute work?**

## WORKSHEET

**1.**There are two types of forces: contact and non-contact.

**a.** Place one tick in each row to show whether the named force is a contact force or a non-contact force.

| Force | Contact | Non-Contact |
|---|---|---|
| Friction | | |
| Air resistance | | |
| Gravitational | | |
| Upthrust | | |
| Magnetic | | |
| Reaction | | |
| Electrostatic | | |

**b.** List three things that forces can do to an object:

1. Change the _______________ of the object.

2. Change the _______________ of the object.

3. Change the _______________ of the object.

## Observe gravitational pull

### ACTIVITY

- **Experiment with Gravity**: Drop a book and a piece of paper from the same height. Observe which falls faster. Crumple the paper and repeat. Discuss the effect of air resistance.

**Project- Activity**:

- Create a poster illustrating different types of forces with examples from everyday life.

### Critical Thinking

- Why do we slide easily on a wet floor but not on a dry one? How does friction play a role in this situation?